Nonlinear Twin-Track Vehicle Model

Mathematical description, parameters, limits, and Python implementation

1 Scope and purpose

This report documents the plant model implemented in `twintrack_model.py`. It follows the structure of the supplied single-track derivation, but describes the actual four-wheel twin-track equations in the code. Controller classes, path tracking, and hardware communication are outside the scope of this document.

The model represents planar translation, yaw motion, and the rotational dynamics of all four wheels. Each wheel has an independent steering angle, drive torque, and brake torque. Tire forces are generated by a simplified Pacejka law and then restricted by a combined-slip grip ellipse.

1.1 Principal assumptions

- The chassis is rigid and planar; heave, pitch, and roll are omitted.
- Vehicle mass is concentrated at the center of gravity and yaw is the only rotational body degree of freedom.
- Each of the four tires is represented separately at its own wheel-center location.
- Normal loads are static. Longitudinal and lateral load transfer are not modeled.
- Aerodynamic drag, suspension compliance, tire relaxation, aligning torque, and actuator dynamics are not modeled.
- All internal quantities use SI units: m, kg, s, rad, N, and N m.

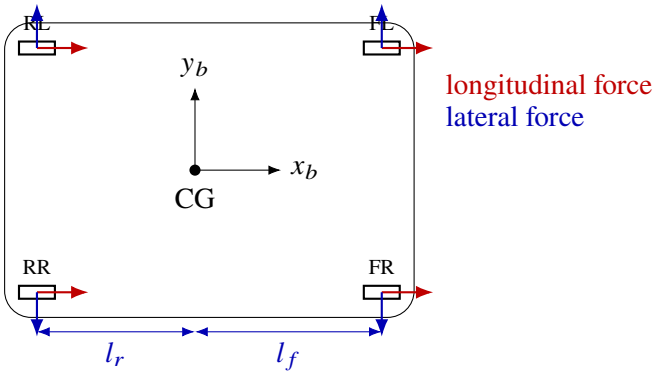


Figure 1: Twin-track geometry and per-wheel tire-force directions.

2 Model structure

At each derivative evaluation, the implementation resolves and clamps inputs, computes the velocity at every wheel, transforms it into the wheel frame, calculates slip variables, evaluates the tire laws, applies the grip ellipse, sums chassis forces and moments, and evaluates wheel acceleration. The resulting continuous derivatives are integrated with a fourth-order Runge–Kutta method.

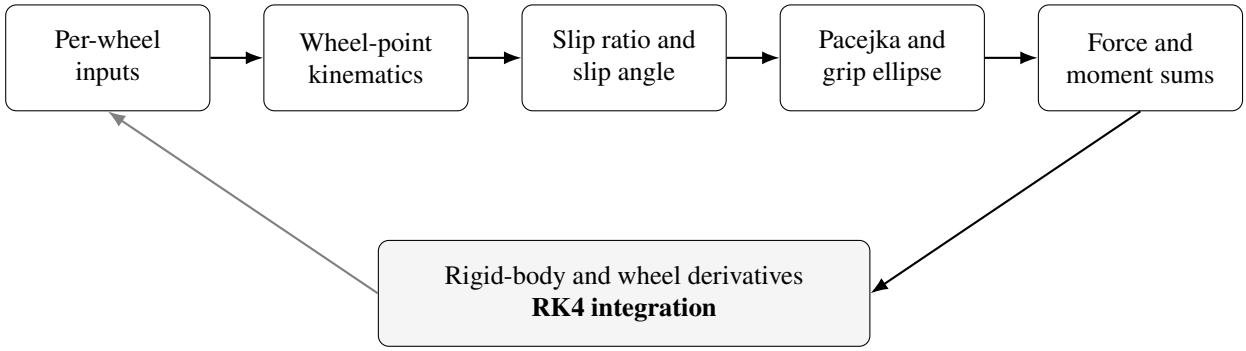


Figure 2: Computational flow used by `TwinTrackModel.derivatives()` and `step()`.

3 State and input spaces

The ten-dimensional state vector is

$$\boldsymbol{x} = \begin{bmatrix} x & y & \psi & v_x & v_y & r & \omega_{\mathrm{FL}} & \omega_{\mathrm{FR}} & \omega_{\mathrm{RL}} & \omega_{\mathrm{RR}} \end{bmatrix}^T . \tag{1}$$

Table 1: States of the implemented plant model.

State	Symbol	Units	Meaning
Global x position	x	m	CG position in the inertial frame
Global y position	y	m	CG position in the inertial frame
Yaw angle	ψ	rad	Body heading in the inertial frame
Longitudinal velocity	v_x	m/s	CG velocity in the body frame
Lateral velocity	v_y	m/s	CG velocity in the body frame
Yaw rate	r	rad/s	Rotation rate about the vertical axis
Wheel angular speeds	$\omega_{\mathrm{FL}}, \omega_{\mathrm{FR}}, \omega_{\mathrm{RL}}, \omega_{\mathrm{RR}}$	rad/s	One rotational state per wheel

The full per-wheel input vector has twelve components:

$$\boldsymbol{u} = \begin{bmatrix} \tau_{d,1\ldots4} & \tau_{b,1\ldots4} & \delta_{1\ldots4} \end{bmatrix}^T . \tag{2}$$

Table 2: Per-wheel inputs.

Input group	Symbols	Units	Sign convention
Drive torques	$\tau_{d,\mathrm{FL}} \ldots \tau_{d,\mathrm{RR}}$	N m	Signed torque applied to each wheel
Brake torques	$\tau_{b,\mathrm{FL}} \ldots \tau_{b,\mathrm{RR}}$	N m	Positive magnitude opposing wheel rotation
Steering angles	$\delta_{\mathrm{FL}} \ldots \delta_{\mathrm{RR}}$	rad	Positive rotation in the wheel/body transformation

The compatibility interface can supply one front steering angle, a total rear-axle drive torque, and axle brake torques. The implementation divides axle totals equally between the corresponding wheels. Explicit per-wheel inputs override these compatibility values.

4 Parameters

Table 3: Geometry, rigid-body, and numerical parameters.

Parameter	Code field	Value	Units
Vehicle mass	`mass`	2.5	kg
Yaw moment of inertia	`moment_of_inertia`	0.027	$\mathrm{kg\,m^2}$
Front axle to CG	`front_axle_len`	0.16	m
Rear axle to CG	`rear_axle_len`	0.14	m
Front track width	`front_track_width`	0.16	m
Rear track width	`rear_track_width`	0.16	m
Front wheel radius	`front_wheel_radius`	0.033	m
Rear wheel radius	`rear_wheel_radius`	0.033	m
Wheel rotational inertia	`wheel_inertia`	0.001	$\mathrm{kg\,m^2}$
Minimum slip speed	`st_min_velocity`	0.2	m/s
Gravity	`gravity`	9.81	$\mathrm{m/s^2}$

Table 4: Tire and limit parameters.

Parameter	Code field	Value	Units
Tire friction coefficient	`tire_friction_coeff`	0.9	–
Rolling resistance coefficient	`rolling_resistance`	0.03	–
Longitudinal Pacejka B	`pacejka_bx`	10.0	–
Longitudinal Pacejka C	`pacejka_cx`	1.9	–
Longitudinal Pacejka E	`pacejka_ex`	0.97	–
Lateral Pacejka B	`pacejka_by`	2.15	–
Lateral Pacejka C	`pacejka_cy`	1.3	–
Lateral Pacejka E	`pacejka_ey`	−1.6	–
Steering limit	`max_steer_deg`	40.0	deg
Front cornering stiffness	`front_cornering_stiff`	16.0	unused
Rear cornering stiffness	`rear_cornering_stiff`	16.0	unused

The two cornering-stiffness fields are retained in the parameter dataclass but are not referenced by the implemented nonlinear force calculation. Lateral stiffness is governed by the Pacejka coefficients and vertical load.

5 Selected log-based calibration

The implemented defaults in the preceding tables remain the baseline values in `twintrack_model.py`. A separate calibration was selected from the 98.04-s recording `session_20260826_145318.mcap`. The optimization used an uninterrupted open-loop rollout initialized from zero state. Rear-current requests and the front steering request were held piecewise constant between logged samples. No state correction or periodic reset was applied.

The current-to-wheel-torque and steering mappings used for the selected candidate are

$$\begin{aligned} \tau_{\mathrm{RL}} &= K_t s_I I_{\mathrm{RL}}, \qquad \tau_{\mathrm{RR}} = -K_t s_I I_{\mathrm{RR}}, \\ \delta &= \frac{\pi}{180}\frac{\delta_{\mathrm{req,deg}}}{g_\delta}, \qquad K_t = 0.01123\ \mathrm{N\,m/A}. \end{aligned} \tag{3}$$

The right-current sign accounts for the opposite right-motor electrical polarity. With $s_I = 2.16355$, the effective fitted conversion is $K_t s_I = 0.024297\ \mathrm{N\,m/A}$.

Table 5: Selected calibration candidate; these values are documented but are not applied to the Python defaults.

Quantity	Symbol or field	Baseline	Selected candidate
Current-to-torque scale	s_I	1.00000	2.16355
Steering conversion gain	g_δ	1.47000	1.32891
Lateral Pacejka stiffness	`pacejka_by`	2.15000	1.19782
Tire friction coefficient	`tire_friction_coeff`	0.90000	1.30819

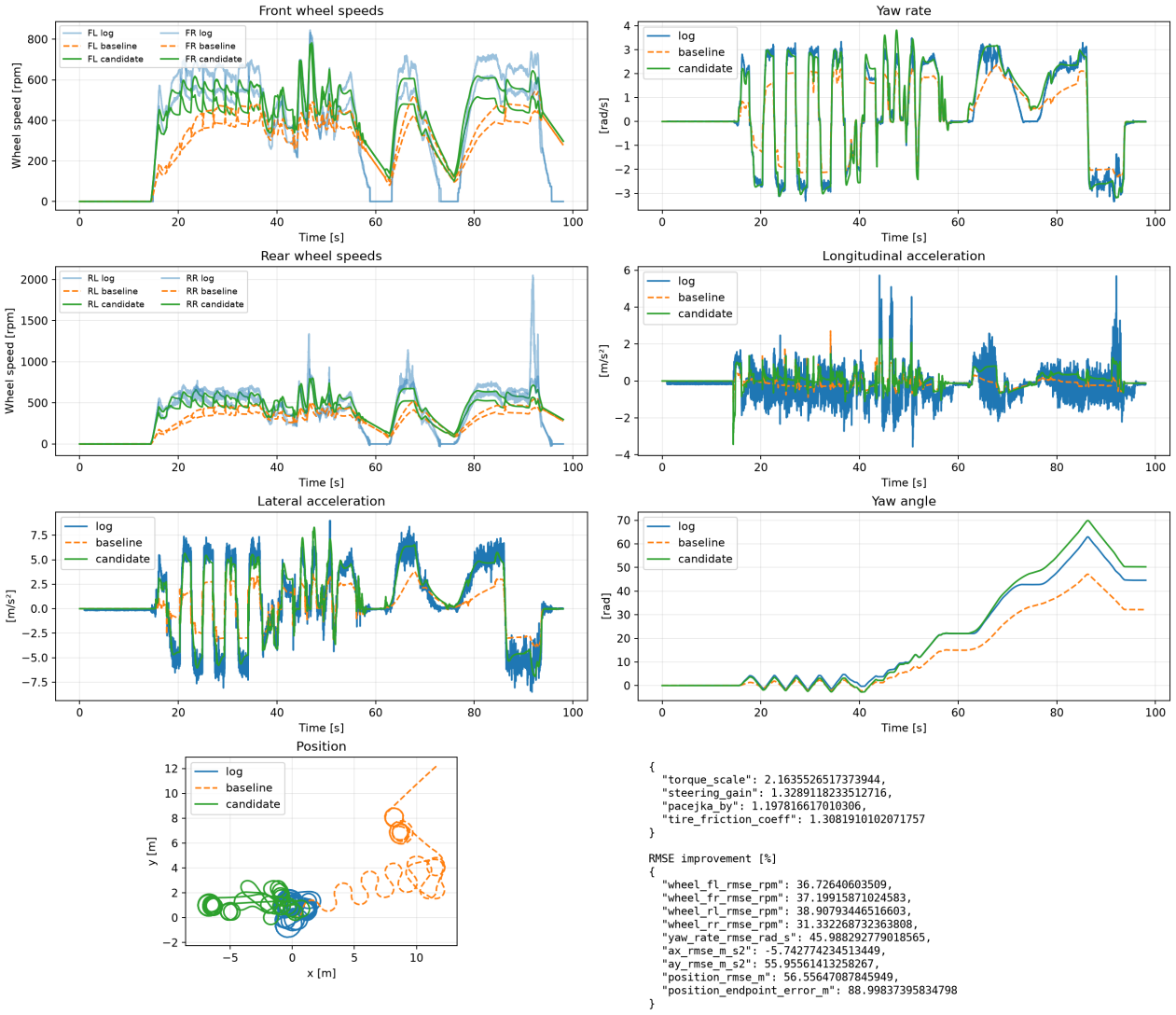
For this vehicle installation, the IMU axes map into the body frame as

$$r = \omega_{z,\mathrm{IMU}}, \qquad a_x = a_{y,\mathrm{IMU}}, \qquad a_y = -a_{x,\mathrm{IMU}}. \tag{4}$$

Table 6: Full-recording open-loop verification errors.

Metric	Baseline RMSE	Candidate RMSE	Improvement
Front-left wheel speed	166.5 rpm	105.4 rpm	36.7%
Front-right wheel speed	176.0 rpm	110.5 rpm	37.2%
Rear-left wheel speed	187.4 rpm	114.5 rpm	38.9%
Rear-right wheel speed	226.5 rpm	155.5 rpm	31.3%
Yaw rate	0.659 rad/s	0.356 rad/s	46.0%
Longitudinal acceleration	0.614 m/s^2	0.649 m/s^2	−5.7%
Lateral acceleration	2.092 m/s^2	0.922 m/s^2	56.0%
Position	9.54 m	4.14 m	56.6%
Final position error	17.34 m	1.91 m	89.0%

The candidate substantially improves wheel speed, yaw rate, lateral acceleration, and accumulated position. Longitudinal-acceleration RMSE changes only slightly and remains below 0.65 m/s^2. Because these quantities were identified jointly from one recording, additional recordings should be used to check parameter repeatability before updating the implemented defaults.




```
{
  "torque_scale": 2.1635526517373944,
  "steering_gain": 1.3289118233512716,
  "pacejka_by": 1.197816617010306,
  "tire_friction_coeff": 1.3081910102071757
}

RMSE improvement [%]
{
  "wheel_fl_rmse_rpm": 36.72640603509,
  "wheel_fr_rmse_rpm": 37.19915871024583,
  "wheel_rl_rmse_rpm": 38.90793446516603,
  "wheel_rr_rmse_rpm": 31.332268732363808,
  "yaw_rate_rmse_rad_s": 45.988292779018565,
  "ax_rmse_m_s2": -5.742774234513449,
  "ay_rmse_m_s2": 55.9556141258267,
  "position_rmse_m": 56.55647087845949,
  "position_endpoint_error_m": 88.99837395834798
}
```


Figure 3: Logged signals, implemented baseline, and the selected calibration candidate over the complete open-loop verification interval.

6 Wheel geometry and kinematics

Wheel-center positions are expressed in the body frame. With front distance l_f, rear distance l_r, front track t_f, and rear track t_r, the implementation uses

$$\begin{aligned} \boldsymbol{p}_{\mathrm{FL}} &= (l_f, +t_f/2), \quad & \boldsymbol{p}_{\mathrm{FR}} &= (l_f, -t_f/2), \\ \boldsymbol{p}_{\mathrm{RL}} &= (-l_r, +t_r/2), \quad & \boldsymbol{p}_{\mathrm{RR}} &= (-l_r, -t_r/2). \end{aligned} \tag{5}$$

For a wheel located at (x_w, y_w), rigid-body translation and yaw give

$$v_{wx,b} = v_x - r y_w, \qquad v_{wy,b} = v_y + r x_w. \tag{6}$$

The velocity is rotated by steering angle δ into the wheel frame:

$$\begin{aligned} v_{\mathrm{long}} &= \cos(\delta)\, v_{wx,b} + \sin(\delta)\, v_{wy,b}, \\ v_{\mathrm{lat}} &= -\sin(\delta)\, v_{wx,b} + \cos(\delta)\, v_{wy,b}. \end{aligned} \tag{7}$$

6.1 Slip variables

For wheel radius R and angular speed ω, the circumferential speed is $R\omega$. A minimum reference speed $v_{\min} = 0.2\,\mathrm{m/s}$ prevents division by zero and excessive numerical sensitivity near rest:

$$v_{\mathrm{ref}} = \max\left(\left|v_{\mathrm{long}}\right|, |R\omega|, v_{\min}\right). \tag{8}$$

$$\kappa = \frac{R\omega - v_{\mathrm{long}}}{v_{\mathrm{ref}}}, \qquad \alpha = \operatorname{atan2}(v_{\mathrm{lat}}, v_{\mathrm{ref}}). \tag{9}$$

The slip-ratio normalization is robust at zero speed, but κ is not explicitly clamped. If wheel and ground speeds have opposite signs, $|\kappa|$ can exceed one.

7 Static normal loads

Normal loads are constant and split equally between left and right tires. For wheelbase $L = l_f + l_r$,

$$F_{z,\text{front axle}} = mg\frac{l_r}{L}, \qquad F_{z,\text{rear axle}} = mg\frac{l_f}{L}. \tag{10}$$

$$F_{z,\mathrm{FL}} = F_{z,\mathrm{FR}} = 5.7225\ \mathrm{N}, \qquad F_{z,\mathrm{RL}} = F_{z,\mathrm{RR}} = 6.5400\ \mathrm{N}. \tag{11}$$

No acceleration-dependent, roll-dependent, or aerodynamic load transfer modifies these values.

8 Simplified Pacejka tire law

The same scalar Magic Formula is used for longitudinal slip κ and lateral slip angle α:

$$\boxed{P(s; B, C, E, D) = D\,\sin(C\,\arctan[Bs - E\,(Bs - \arctan(Bs))])}. \tag{12}$$

$$\begin{aligned} F_{x,\mathrm{raw}} &= P(\kappa; B_x, C_x, E_x, \mu F_z), \\ F_{y,\mathrm{raw}} &= -P(\alpha; B_y, C_y, E_y, \mu F_z). \end{aligned} \tag{13}$$

The negative sign in the lateral equation makes the generated force oppose positive lateral slip. Peak force D is not an independent parameter; it is set to μF_z for both directions.

9 Combined-slip grip ellipse

The implementation contains a tire grip ellipse. Raw longitudinal and lateral tire forces are tested against

$$e = \left(\frac{F_x}{F_{x,\max}}\right)^2 + \left(\frac{F_y}{F_{y,\max}}\right)^2. \tag{14}$$

If $e \leq 1$, the raw forces are retained. If $e > 1$, both components are scaled in their current direction:

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = e^{-1/2} \begin{bmatrix} F_{x,\mathrm{raw}} \\ F_{y,\mathrm{raw}} \end{bmatrix}. \tag{15}$$

The implementation sets $F_{x,\max} = F_{y,\max} = \mu F_z$. Consequently, the configured ellipse is circular in normalized force coordinates. With the default parameters, the limits are 5.150 N per front tire and 5.886 N per rear tire.

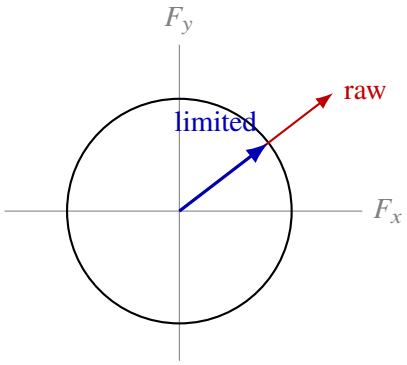


Figure 4: Radial scaling of a raw combined tire-force vector to the grip boundary.

10 Wheel rotational dynamics

For each wheel, drive torque, brake torque, longitudinal tire force, and rolling resistance determine angular acceleration:

$$\dot{\omega} = \frac{\tau_d - \operatorname{sgn}(\omega)\tau_b - RF_x - C_{rr}F_zR\operatorname{sgn}(\omega)}{I_w}. \tag{16}$$

Brake torque is treated as a non-negative magnitude opposing the current wheel direction. Rolling resistance is implemented as a wheel torque proportional to normal force and radius. At exactly zero angular velocity both sign functions return zero.

11 Force projection and chassis dynamics

Tire-frame forces are rotated back to the body frame for each wheel:

$$\begin{aligned} F_{x,b} &= \cos(\delta)F_x - \sin(\delta)F_y, \\ F_{y,b} &= \sin(\delta)F_x + \cos(\delta)F_y. \end{aligned} \tag{17}$$

The force and yaw-moment sums are

$$\sum F_x = \sum_i F_{x,b,i}, \qquad \sum F_y = \sum_i F_{y,b,i}, \qquad M_z = \sum_i \left(x_i F_{y,b,i} - y_i F_{x,b,i}\right). \tag{18}$$

The global-position and body-velocity derivatives are

$$\begin{aligned} \dot{x} &= v_x \cos(\psi) - v_y \sin(\psi), \\ \dot{y} &= v_x \sin(\psi) + v_y \cos(\psi), \\ \dot{\psi} &= r, \end{aligned} \tag{19}$$

$$\dot{v}_x = \frac{\sum F_x}{m} + v_y r, \qquad \dot{v}_y = \frac{\sum F_y}{m} - v_x r, \qquad \dot{r} = \frac{M_z}{I_z}. \tag{20}$$

12 Numerical integration

The method `step()` advances all ten states using classical explicit RK4:

$$k_1 = f(x, u), \tag{21}$$

$$k_2 = f\left(x + \frac{\Delta t}{2} k_1, u\right), \tag{22}$$

$$k_3 = f\left(x + \frac{\Delta t}{2} k_2, u\right), \tag{23}$$

$$k_4 = f\left(x + \Delta t\, k_3, u\right), \tag{24}$$

$$x_{k+1} = x_k + \frac{\Delta t}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right). \tag{25}$$

Inputs are held constant across the four RK4 derivative evaluations. The caller is responsible for selecting a sufficiently small positive time step. The model does not contain an adaptive integrator, event handling, or state projection.

13 Implementation map

Implementation element	Location	Responsibility
`TwinTrackParams`	lines 19–47	Immutable defaults for geometry, mass, tires, limits, and environment
`TwinTrackState`	lines 50–61	Ten continuous plant states
`TwinTrackInput`	lines 73–92	Twelve independent per-wheel inputs
`TwinTrackModel.derivatives`	lines 222–398	Continuous dynamics and diagnostic wheel forces
`TwinTrackModel.step`	lines 400–514	One fixed-step RK4 update
`_resolve_inputs`	lines 516–571	Compatibility mapping, overrides, and steering clamp
`pacejka_force`	lines 574–578	Scalar simplified Magic Formula
`traction_ellipse_limit`	lines 581–592	Combined-slip radial force scaling

Line numbers refer to the documented version of `twintrack_model.py` at the time this report was generated.

14 Limits and saturation

Quantity	Implemented behavior	Default
Steering angle	Hard clamp applied independently to all four wheel angles	$\pm 40^\circ$
Drive torque	No saturation in the plant model	unbounded input
Brake torque	No magnitude saturation; sign follows wheel rotation	unbounded magnitude
Tire force	Pacejka peak plus combined-slip grip boundary	$\mu = 0.9$
Slip denominator	Lower bounded for numerical regularization	0.2 m/s
Integrator step	Caller-selected fixed RK4 step	no built-in default

The plant model does not saturate torque commands. Any current limit, current-to-torque conversion, motor speed limit, command delay, or actuator rate limit must be applied outside the model or added as an actuator submodel. The tire grip ellipse limits the resulting tire forces; it is not an electrical input clamp.

15 Outputs and diagnostic quantities

The derivative function returns the state derivative and a dictionary of per-wheel diagnostic values. For FL, FR, RL, and RR it reports tire-frame F_x, tire-frame F_y, static normal force F_z, slip ratio κ, and slip angle α.

15.1 Coordinate and sign conventions

- Body x points forward and body y follows the yaw-kinematics convention.
- Wheel positions use positive y for the left side and negative y for the right side.
- Positive wheel angular speed gives positive circumferential speed $R\omega$.
- Positive drive torque increases wheel speed; positive brake magnitude opposes current wheel rotation.
- Right-motor electrical polarity is not handled by the plant. Hardware-topic signs must be normalized before supplying torque.
- For the verified IMU mounting, body longitudinal acceleration is sensor y, while body lateral acceleration is negative sensor x.

16 Known limitations and identification priorities

- Static normal loads cannot reproduce longitudinal or lateral load transfer.
- A single friction coefficient and identical front/rear Pacejka shape families restrict tire asymmetry.
- No steering ratio, steering offset, actuator delay, or steering time constant is included.
- No motor-current dynamics, torque saturation, drivetrain efficiency, back-EMF, or differential compliance is included.
- No aerodynamic drag is applied to the body.
- The rolling-resistance sign law is discontinuous at zero wheel speed.
- The cornering-stiffness parameters currently have no effect.

For log-based identification, actuator mappings should be estimated first: current request → measured current → wheel torque, and steering request → effective road-wheel angle. Tire coefficients should then be fitted on short windows initialized from measured pose, body velocity, yaw rate, and all four wheel speeds. Long continuous position replay is best reserved for final validation because small yaw errors accumulate rapidly.

17 Minimal implementation usage

1. Construct `TwinTrackParams` and `TwinTrackModel`.
2. Initialize all ten values in `TwinTrackState`.
3. Build `TwinTrackInput` with torques in N m and steering in radians.
4. Call `model.step(state, dt, wheel_inputs=input)`.
5. Use `model.derivatives(...)` when force and slip diagnostics are required.

18 References

References

[1] H. B. Pacejka, *Tire and Vehicle Dynamics*. Elsevier.

[2] D. Schramm, M. Hiller, and R. Bardini, *Vehicle Dynamics: Modeling and Simulation*. Springer.

[3] D. Efremov, *Nonlinear Single-Track Model*, supplied `ModelDerivation.pdf`, 2020.

[4] Project implementation: `twintrack_model.py`, including `TwinTrackModel`, `TwinTrackParams`, `TwinTrackState`, and `TwinTrackInput`.

[5] Project recording: `session_20260826_145318.mcap`, open-loop model verification and parameter calibration, 98.04 s.